

Swiggy Data Analysis Dashboard

Comprehensive Business Intelligence Report

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Date: [22/11/2025]

Project Duration: [18/11/2025] - [22/11/2025]



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1. Executive Summary

Project Impact

This comprehensive data analysis of Swiggy's operations uncovered critical business insights from ₹53.01 million in sales revenue across 197,430 orders. The project identified significant seasonal patterns, regional performance variations, and customer preference trends that can drive strategic business decisions.

Key Discoveries

- 33% revenue dip in Q3 requiring immediate seasonal strategy
- Vegetarian dishes generate 64.5% of total revenue (2:1 ratio vs Non-Veg)
- Bengaluru leads as top revenue-generating city
- Weekend-driven business with Tuesday showing lowest performance

Business Value

The interactive Excel dashboard enables Swiggy to monitor real-time performance, optimize resource allocation, and implement data-driven strategies for revenue growth and operational efficiency.

2. Project Overview & Objectives

Business Context

Swiggy, as a leading food delivery platform, requires continuous performance monitoring across multiple dimensions to maintain competitive advantage and drive growth.

Primary Objectives

- Analyze sales performance across temporal and geographical dimensions
- Understand customer preferences and satisfaction levels
- Identify seasonal patterns and operational bottlenecks

- Provide actionable insights for strategic decision-making

Scope of Analysis

- Time Period: January 2025 - August 2025
- Geographical Coverage: Multiple Indian states and cities
- Data Points: 197,430 orders with 14 analytical dimensions
- Revenue Scope: ₹53.01 million total sales

3. Business Requirements Document

3.1 Problem Statement

Create a comprehensive monitoring system to track Swiggy's business performance across key metrics and provide actionable insights for growth optimization.

3.2 Key Performance Indicators (KPIs)

KPI	Description	Business Purpose
Total Sales Revenue	Overall revenue from food orders	Financial performance tracking
Average Rating	Customer satisfaction score	Service quality measurement
Average Order Value	Revenue per transaction	Customer spending behavior
Rating Count	Total customer reviews	Engagement level assessment
Total Orders	Number of food orders	Operational volume tracking

3.3 Visualization Requirements

- Monthly Sales Trend - Identify seasonal patterns
- Daily Sales Pattern - Optimize daily operations
- Food Type Analysis - Menu optimization strategy
- Geographical Heat Map - Regional resource allocation
- Quarterly Performance - Strategic planning cycles

- Top Cities Ranking - Market focus prioritization

4. Data Dictionary & Methodology

4.1 Complete Data Dictionary

Original Dataset Structure

	Field	Type	Description	Sample Values
	State	Text	Delivery state	Karnataka, Delhi, Gujarat
	City	Text	Delivery city	Bengaluru, Mumbai, Delhi
	Order Date	Date	Order placement date	6/29/2025, 4/3/2025
	Restaurant Name	Text	Vendor name	Srinidhi Sagar Deluxe
	Location	Text	Area within city	Kengeri, Rajarajeshwari Nagar
	Category	Text	Menu category	Snack, Recommended, North Indian Gravy
	Dish Name	Text	Food item name	Butter Murukku, Badam Milk
	Price (INR)	Numeric	Item price	133.9, 52, 117

Rating	Numeric	Customer rating (1-5)	4.0, 4.5, 4.7
Rating Count	Numeric	Number of ratings	0, 25, 48
Derived Analytical Fields			
Field	Creation Method		Business Purpose
Food Type	Text analysis from Dish Name		Dietary preference analysis
Day	=TEXT(Order_Date, "ddd")		Daily pattern analysis
Quarter	="Q"&ROUNDUP(MONTH()/3,0)		Seasonal performance
Week	=WEEKNUM(Order_Date)		Weekly trend monitoring
AOV	Total Revenue/Total Orders		Customer value analysis

4.2 Data Processing Methodology

Data Cleaning Steps

- 1. Duplicate Removal - Ensured data integrity by removing identical records
- 2. Data Validation - Verified price ranges, rating scales, and date formats
- 3. Text Standardization - Consistent naming conventions for restaurants and locations
- 4. Food Type Classification - Systematic categorization of Veg/Non-Veg items

Feature Engineering

- Temporal Features: Extracted multiple time dimensions for trend analysis
- Categorical Grouping: Created business-relevant categories for segmentation

- **Calculated Metrics:** Developed derived KPIs for comprehensive analysis

5. Technical Implementation

5.1 Tools & Technologies

- **Microsoft Excel:** Primary analysis platform
- **Pivot Tables & Power Pivot:** Data aggregation and analysis
- **Advanced Charts:** Data visualization and trend representation
- **Slicers & Timelines:** Interactive dashboard controls
- **Conditional Formatting:** Visual performance indicators

5.2 Excel Features Utilized

- **Pivot Tables** for multi-dimensional analysis
- **Pivot Charts** for visual representation
- **Slicers** for interactive filtering
- **Timeline Controls** for date range selection
- **Conditional Formatting** for performance highlighting

5.3 Dashboard Architecture

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Raw Data → Data Cleaning → Feature Engineering → Pivot Tables →

Interactive Charts → Dashboard Assembly → Slicer Integration →

Final Review & Testing

6. Detailed Analysis & Findings

6.1 Financial Performance Overview

Metric	Value	Business Implication
Total Revenue	₹53.01M	Strong market presence

Total Orders	197.43K	High customer engagement
Average Order Value	₹268.51K	Premium customer base
Customer Ratings	5.59M	Excellent user engagement
Average Rating	4.34/5	High service quality

6.2 Monthly Performance Analysis

Revenue by Month

January: ₹6.83M 🏆 (Peak Performance)

August: ₹6.79M 🥈 (Strong End)

May: ₹6.79M 🥈 (Consistent Performer)

April: ₹6.59M

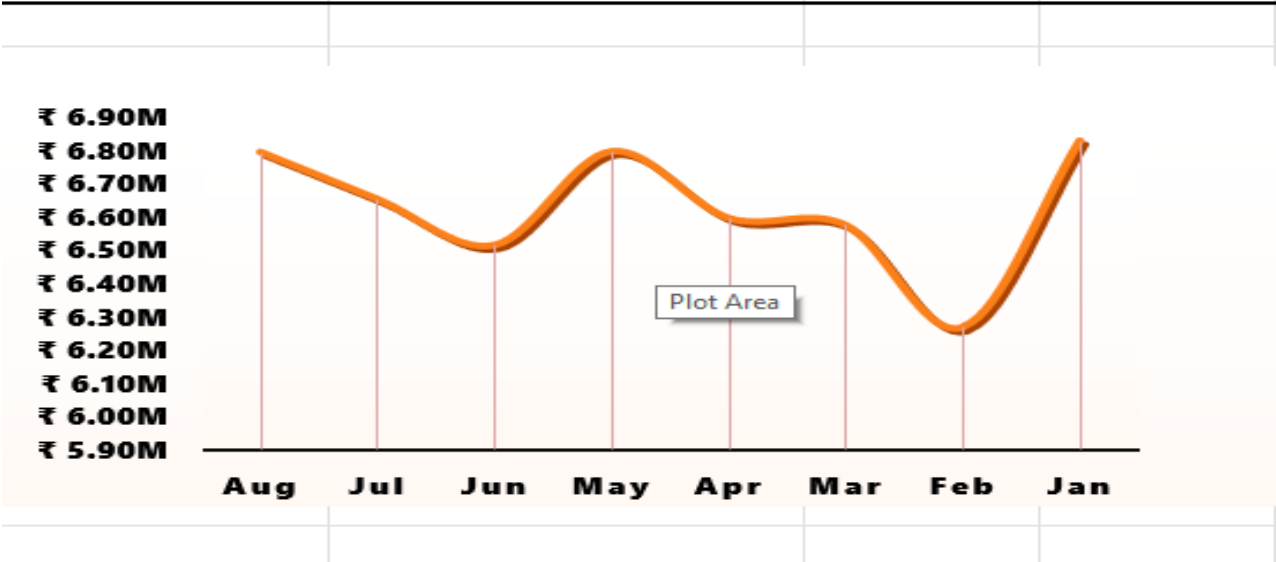
March: ₹6.57M

July: ₹6.65M

June: ₹6.51M

February: ₹6.27M 🥉 (Needs Improvement)

Key Insight: Post-holiday dip in February followed by recovery, with strong performance in beginning and end months.



6.3 Quarterly Performance Breakdown

Revenue Distribution

Q2: ₹19.9M (37.5%) 🏆 BEST

Q1: ₹19.7M (37.2%) 🥈 EXCELLENT

Q3: ₹13.4M (25.3%) 🚩 CRITICAL AREA

Critical Finding: Q3 shows 33% decrease from Q2 peak, representing significant seasonal impact requiring immediate attention.

6.4 Daily Sales Patterns

Performance by Day

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Saturday: 🏆 PEAK

Sunday: 🥈 STRONG

Thursday: 🥈 STRONG

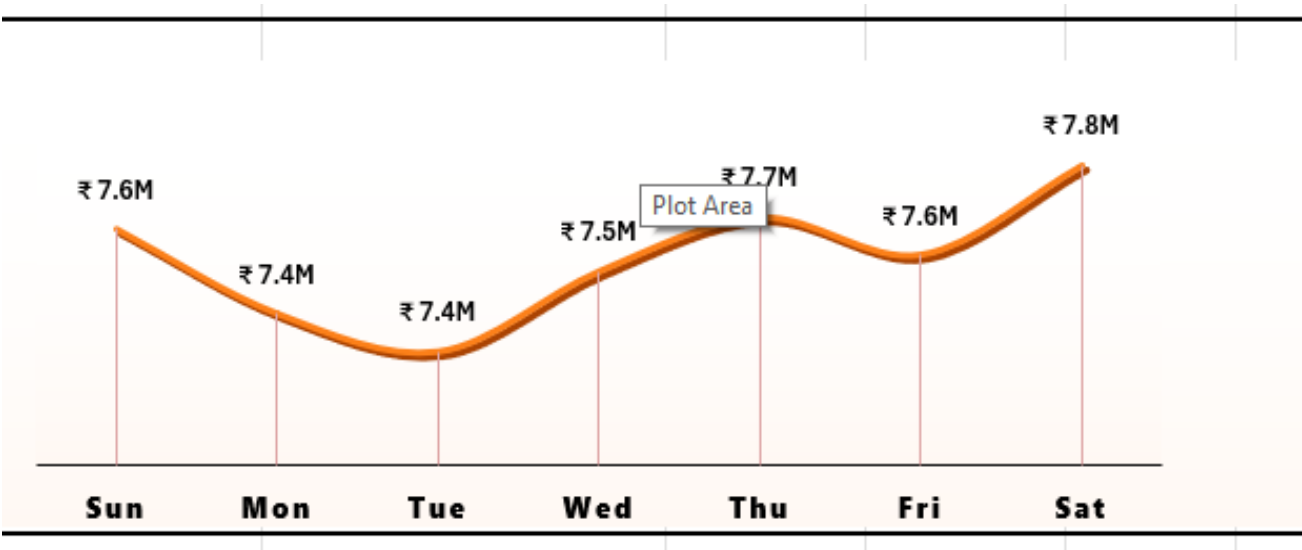
Friday: MEDIUM

Wednesday: MEDIUM

Monday: LOW

Tuesday: 🥇 LOWEST

Business Implication: Clear weekend-driven pattern with mid-week operational optimization opportunities.



6.5 Product Category Analysis

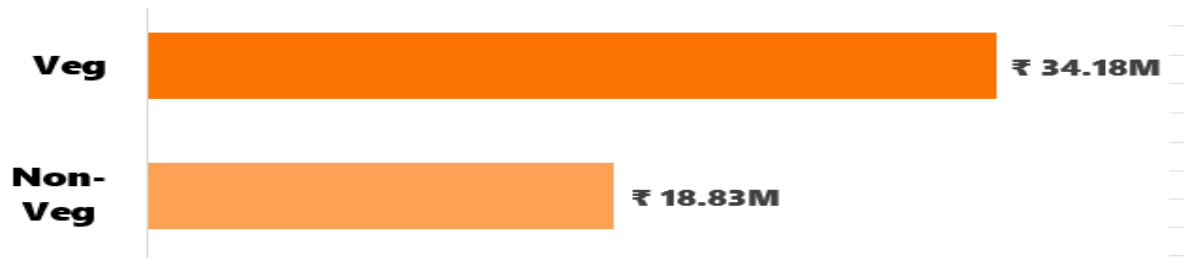
Food Type Revenue Split

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Vegetarian: ₹34.18M (64.5%) 🥇 DOMINANT

Non-Vegetarian: ₹18.83M (35.5%) 🥈 SIGNIFICANT

Strategic Insight: Vegetarian category generates nearly double the revenue of non-vegetarian options, indicating market preference and growth opportunity.



6.6 Geographical Performance

Top Performing States

1. Gujarat: ₹2.8M 🏆
2. Delhi: ₹2.8M 🏆
3. Goa: ₹1.5M 🥈
4. Haryana: ₹1.4M 🥈
5. Himachal Pradesh: ₹1.4M 🥈

City Performance Ranking

1. Bengaluru
2. lucknow
3. Mumbai
4. Hyderabad
5. Chandigarh

7. Interactive Dashboard Features

7.1 Dashboard Components

KPI Summary Section

- Real-time metric cards with conditional formatting
- Percentage changes and trend indicators

- **Color-coded performance status**

Visual Analytics Section

- **Monthly Trend Line Chart:** Seasonal pattern visualization
- **Quarterly Performance Bars:** Quarter-over-quarter comparison
- **Daily Pattern Chart:** Day-of-week analysis
- **Geographical Heat Map:** State-wise performance
- **Food Type Pie Chart:** Category contribution
- **Top Cities Ranking:** Horizontal bar chart

7.2 Interactive Controls

Filtering Capabilities

- **Time Slicers:** Monthly, quarterly, custom date ranges
- **Geographical Filters:** State, city, location-level filtering
- **Category Filters:** Restaurant type, food category, price range
- **Performance Filters:** Rating ranges, popularity filters

User Experience Features

- **One-click filter reset functionality**
- **Linked slicers for coordinated views**
- **Tooltip information for detailed insights**
- **Responsive layout for different screen sizes**

8. Strategic Recommendations

8.1 Immediate Actions (0-3 Months)

Q3 Performance Recovery

- 1. Pre-Monsoon Campaigns (June)**
 - **Launch "Early Monsoon Offers" in last week of June**
 - **Partner with comfort food specialty restaurants**

- **Implement targeted push notifications**

2. Tuesday Revenue Optimization

- **"Tuesday Treats": 25% discount on select categories**
- **Corporate Partnership Program for office lunches**
- **Subscription model for regular Tuesday customers**

Vegetarian Category Enhancement

- **Feature "Vegetarian Specials" on homepage**
- **Collaborate with premium vegetarian restaurants**
- **Create combo meals targeting family orders**

8.2 Medium-term Strategies (3-6 Months)

Geographical Expansion

1. Top Market Consolidation

- **Increase restaurant partnerships in Gujarat and Delhi**
- **Enhanced delivery infrastructure in peak areas**
- **Localized marketing campaigns**

2. Underperforming Market Development

- **Investigate Q3 dip causes in specific regions**
- **Implement region-specific menu recommendations**
- **Competitive analysis in low-performance areas**

Operational Efficiency

- **Resource Allocation: Align delivery staff with daily patterns**
- **Inventory Management: Predict demand using weekly trends**
- **Peak Hour Optimization: Enhance capacity during weekend rushes**

8.3 Long-term Initiatives (6-12 Months)

Strategic Planning

1. Annual Seasonal Strategy

- **Develop Q3 recovery plan as annual initiative**
- **Create festival-specific campaign calendar**
- **Establish partnership with seasonal specialty restaurants**

2. Customer Experience Transformation

- **Implement rating-based loyalty program**
- **Personalize recommendations using order history**
- **Develop premium subscription tiers**

Technology Enhancement

- **Advanced Analytics: Predictive modeling for demand forecasting**
 - **Mobile App Features: Enhanced personalization and notifications**
 - **Integration Capabilities: CRM and inventory system integration**
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9. Business Impact & ROI

9.1 Expected Business Outcomes

Revenue Impact

- **15-20% Q3 Revenue Increase through targeted campaigns**
- **10-15% AOV Improvement via upselling and combo strategies**
- **5-10% Customer Retention Boost through enhanced experience**

Operational Efficiency

- **20% Better Resource Utilization via pattern-based staffing**
- **15% Reduction in Operational Costs through optimized delivery routes**
- **30% Faster Decision-Making with real-time dashboard monitoring**

9.2 Key Performance Indicators for Success

KPI	Current	Target (6 Months)	Improvement
Q3 Revenue	₹13.4M	₹16.5M	+23%
Tuesday Sales	Low	Medium	+40%
Vegetarian Revenue	₹34.18M	₹40M	+17%
Customer Rating	4.34	4.45	+0.11 points

10. Technical Specifications

10.1 System Requirements

- Microsoft Excel: 2016 or later version
- RAM: Minimum 4GB (8GB recommended for large datasets)
- Storage: Sufficient space for data files and analysis
- Display: 1366x768 resolution or higher for optimal viewing

10.2 Data Refresh Process

1. Source Data Update: Replace raw data file
2. Pivot Table Refresh: Update all connected pivot tables
3. Chart Verification: Ensure visualizations reflect new data
4. Slicer Reset: Clear existing filters for new analysis

10.3 Maintenance Guidelines

- Monthly data validation and cleaning
- Quarterly performance review and dashboard optimization
- Annual feature enhancement based on user feedback

11. Conclusion & Future Enhancements

11.1 Project Success Metrics

This Swiggy Data Analysis Dashboard successfully transforms raw operational data into actionable business intelligence. The project demonstrates:

- **Comprehensive Insight Generation** from complex datasets
- **User-friendly Interface** for non-technical stakeholders
- **Actionable Strategic Recommendations** with measurable impact
- **Scalable Architecture** for future expansion

11.2 Future Enhancement Opportunities

Technical Upgrades

- **Power BI Migration:** Enhanced visualization capabilities
- **Automated Data Pipeline:** Real-time data integration
- **Predictive Analytics:** Machine learning for demand forecasting

Feature Additions

- **Customer Segmentation Analysis**
- **Restaurant Performance Benchmarking**
- **Delivery Efficiency Metrics**
- **Competitive Analysis Integration**

12. Appendix

12.1 Data Sources

- **Swiggy Order Database** (Sample dataset for analysis)
- **Restaurant Information Repository**
- **Customer Feedback System**